School of Computer Science Engineering and Technology

Course- B.Tech.	Type- Core
Course Code- **2025CSET106**	Course Name- **Discrete Mathematical Structures**
Year- I	Semester-II

Tutorial-1

Tutorial No.	Name	CO1	CO2	CO3
1	**Propositions, Logical Expressions**	✓	--	--

Objective: The main objective of this tutorial is to learn about propositions, basic operations in propositions, translate English statements into logical expressions.

Q1. Consider the following six statements:

a. Ice floats in water. **T**
b. 7>26. **F**
c. How far is Delhi from Greater Noida?
d. 2 + 2 = 5 **T**
e. Oh no! -
f. It is false that 2 + 2 = 5. **T**
g. x is a prime number -
h. Tomorrow will be a great day. -

Recognize the propositional statements. Also, state their truth values.

A proposition must be declarative and either true or false (but not both) whereas anything that is a question, command, emotion, variable, undetermined or ambiguous statement is not a proposition.

Q2. Suppose that during the last year, the average daily passenger counts and on-time performance of three metro lines were as follows:

- Red Line carried 1.8 million passengers per day and had 92% on-time performance.
- Blue Line carried 1.2 million passengers per day and had 88% on-time performance.
- Green Line carried 1.5 million passengers per day and had 95% on-time performance.

Determine the truth value of each of the following propositions.

a. The Red Line carried the highest number of passengers, and the Green Line had the best on-time performance. **T**
b. The Blue Line carried the fewest passengers and either the Red Line or the Green Line had on-time performance above 90%. **T**
c. The Green Line did not carry the highest number of passengers, and neither the Red Line nor the Blue Line had the worst on-time performance. **F**

Q3**.** For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.

a. Experience with C++ or Java is required. **Inclusive-OR**
b. Dinner includes rice or bread. **Exclusive-OR**
c. Flights are cancelled if visibility is poor or if wind speed exceeds 80 km/h.. **Inclusive-OR**
d. When you enroll in the new internet plan, you receive a free router or a 3-month subscription discount. **Exclusive-OR**

Q3. Determine whether the given compound propositions are tautology, contradiction or contingent

a. $(p \vee q) \wedge \neg (p \vee q)$ **Contradiction**

b. $(p \wedge q) \vee \neg r$ **Contingent**

c. $(p \oplus q) \vee \neg (p \oplus q)$ **Tautology**

Q4. Let p and q be the propositions

p: The library is open. q: The librarian is available.

Write the following propositions using p, q, and logical connectives (including negations).

a. The library is open but the librarian is not available. **$p \wedge \neg q$**
b. The library is not open, and the librarian is not available. **$\neg p \wedge \neg q$**
c. Either the library is open or the librarian is available (or both). **$p \vee q$**

Write the following propositions using p, q, and logical connectives (including negations).

Q5. Let p, q, and r be the propositions:

- p : You pass the fitness test.
- q : You attend all training sessions.
- r : You are selected for the team.

Write each statement using p**,** q**,** r**,** and basic logical connectives.

a. You are selected for the team, but you do not attend all training sessions. **$r \wedge \neg q$**
b**.** You pass the fitness test, attend all training sessions, and are selected for the team**.** **$p \wedge q \wedge r$**
c. You pass the fitness test, but you do not attend all training sessions; nevertheless, you are selected for the team**.** **$p \wedge \neg q \wedge r$**

Q6. Let p, q, and r be the propositions

p : You have the flu.
q : You miss the final examination.
r : You pass the course.
Express each of these propositions as an English sentence.

a. $p \vee q \vee r$

You have the flu or you miss the final examination or you pass the course.

b. $\neg p \wedge q$

You don't have the flu and you miss the final examination

c. $(p \wedge q) \vee (\neg q \wedge r)$

You have the flu and you miss the final examination or you don't miss the final examination and you pass the course.